

SDSC6024: DISSERTATION FOR ARTIFICIAL INTELLIGENCE FOR SCIENCES

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Dissertation for Artificial Intelligence for Sciences

Subject Code

SDSC - Data Science

Course Number

6024

Academic Unit

Data Science (DS)

College/School

College of Computing (CC)

Course Duration

Two Semesters

Credit Units

0-6

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

NIL

Precursors

NIL

Equivalent Courses

DSC6024 Dissertation for Artificial Intelligence for Sciences

Exclusive Courses

DSC6023 Internship in Artificial Intelligence for Sciences

SDSC6023 Internship in Artificial Intelligence for Sciences

Part II Course Details

Abstract

This course aims to develop a student's expertise in his/her chosen subject area through application of AI for science knowledge and techniques, and demonstrate a student's ability to present and organize subject knowledge scholarly in an independent research.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Carry out independent study for problem solving and solution seeking.	20	x	x	
2	Integrate AI knowledge and techniques of diverse subjects in formulating and analyzing scientific problems.	20	x	x	
3	Assess critically appropriateness of research methodology and strategies in approaching problems.	20	x	x	
4	Interpret insights and novel findings in conducted data analytics studies.	20	x		x
5	Prepare a coherent dissertation with effective presentation of literature and analysis of results.	20		x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Consultation	Students will meet with their chosen supervisors to identify appropriate project topics and make research plans.	1, 2, 3, 4	13 hours/sem

2	Individual works	Students will conduct the selected research project under the guidance of their supervisor.	2, 3, 4, 5	26 hours/sem
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Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Project Proposal	3, 4	20	-	Yes
2	Project Milestone Meetings	1, 2, 3, 4	20	-	Yes
3	Dissertation	1, 2, 3, 4, 5	40	-	Yes
4	Oral Presentation	5	20	-	Yes

Continuous Assessment (%)

100

Assessment Rubrics (AR)**Assessment Task**

Project Proposal (for students admitted in Semester A 2026/27 & thereafter)

Criterion

20%

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project Milestone Meetings (for students admitted in Semester A 2026/27 & thereafter)

Criterion

20%

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Dissertation (for students admitted in Semester A 2026/27 & thereafter)

Criterion

40%

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Oral Presentation (for students admitted in Semester A 2026/27 & thereafter)

Criterion

20%

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

No formal syllabus. Students will be required to undertake individually supervised research and dissertation preparation.

Reading List